Curriculum Vitae

Charikleia Karageorgiou

Postdoctoral Researcher, Department of Biological Sciences, University at Buffalo

✉charikle@buffalo.edu; clio.karageorgiou@gmail.com

Google Scholar | ORCID | Bluesky | Website

EDUCATION

Sept 2017 – Sept 2022	PhD *cum laude*, Evolutionary Genetics, Universitat Autònoma de Barcelona, Spain.
Sept 2016 – July 2017	MSc Bioinformatics, Universitat Autònoma de Barcelona, Spain.
Sept 2014 – July 2015	MSc Advanced Genetics, Universitat Autònoma de Barcelona, Spain.
Sept 2011 – June 2014	BSc Biomedicine, University of East Anglia, United Kingdom.

ACADEMIC POSITIONS

Dec 2026 – Present	Tenure-track Assistant Professor, BiRC, Department of Molecular Biology and Genetics, Aarhus University, Denmark.
Jan 2023 – Nov 2026	Postdoctoral Researcher, Department. Biological Sciences, University at Buffalo, SUNY, NY, USA. Mentor: Dr. Omer Gokcumen
Dec 2021 – July 2022	Research Fellow, Institute of Computer Science, Foundation for Research and Technology Hellas - Greece. Supervisor: Dr. Pavlos Pavlidis
July 2015 – Dec 2016	Research Assistant, Grup de Genòmica, Bioinformàtica i Biologia Evolutiva, Universitat Autònoma de Barcelona, Spain.

VISITING POSITIONS & INTERNSHIPS

Oct 2025 – Nov 2025	Visiting Researcher, Center for Interdisciplinary Theoretical and Mathematical Sciences, RIKEN, Japan. Mentor: Dr. Leo Speidel
Aug 2021 – Feb 2022	Visiting PhD Researcher, Institute of Molecular Biology and Biotechnology, Foundation for Research and Technology Hellas, Greece. Funded by Marie Skłodowska-Curie Actions, Research and Innovation Staff Exchange (RISE). Supervisor: Dr. John Vontas
Feb 2021 – June 2021	PhD Internship, University of Zürich, Switzerland. Supervisor: Dr. Martin Kapun

HONORS & AWARDS

2025	DeLill Nasser Award for Professional Development in Genetics, Genetics Society of America. ($1,000)
2022	*Cum Laude*, PhD Defense, Universitat Autònoma de Barcelona, Spain.
2020	Swiss-European Mobility Programme (SEMP) Grant (440 CHF/month, declined due to COVID-19 pandemic)
2017 – 2022	PIF PhD fellowship from the Universitat Autònoma de Barcelona, Spain. (~75,000 €)

PUBLICATIONS & PREPRINTS

* denotes co-first authorship, # denotes corresponding author

12. Aguado, W.D., **Karageorgiou, C.**, Kassem, S., Hernandez Rojas, R., Veilleux, C. C., Melin A. D., Vogel, E. R., Ruhl, S., Gokcumen, O.#, & Pajic, P.#, 2026. Multiscale structural evolution of proline-rich protein genes shapes primate salivary defenses. *Molecular Biology and Evolution*, msag224. [doi: https://doi.org/10.1093/molbev/msag224]

11. **Karageorgiou, C.**, Pajic, P., Ruhl, S., & Gokcumen, O.#, 2026. Convergent evolution through independent rearrangements in the primate amylase locus. *Cell Reports*, *45*(6), 117445. [doi: https://doi.org/10.1016/j.celrep.2026.117445]

10. Scheer, K.*, Landau, L. J. B.*, Jorgensen, K., **Karageorgiou, C.**, Siao, L., Alkan, C., Morales Rivera, A. M., Osborne, C., Garcia, O. A., Pearson, L., Kiyamu, M., Rivera-Ch, M., León-Velarde, F., Lee, F. S., Brutsaert, T., Bigham, A. W.#, & Gokcumen, O.#, 2026. Rapid adaptive increase of amylase gene copy number in Indigenous Andeans. *Nature Communications*, *17*(1), 3822. [doi: https://doi.org/10.1038/s41467-026-71450-8]

9. **Karageorgiou, C.**#, Leffler, E. M.#, Dennis, M. Y.#, & Gokcumen, O.#, 2026. Structural variation in context: mechanisms, functions and selection regimes across the tree of life. *Heredity*, *135*(3), 121–125. [doi: https://doi.org/10.1038/s41437-026-00832-x]

8. Acharjee, M.*, Humphrey, K.*, Lactaoen, V., **Karageorgiou, C.**, Liu, B., Gokcumen, O., & Rusche, L. N.#, 2026. The deacetylase Sir2 is the primary sensor driving transcriptional changes in response to low NAD+ in the yeast Kluyveromyces lactis. *Genetics*, *232*(4), iyag025. [doi: https://doi.org/10.1093/genetics/iyag025]

7. Yilmaz, F.*, **Karageorgiou, C.***, Kim, K.*, Pajic, P., Scheer, K., Human Genome Structural Variation Consortium, Beck, C. R., Torregrossa, A.-M., Lee, C.#, & Gokcumen, O.#, 2024. Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation. *Science*, *386*(6724), eadn0609. [doi: https://doi.org/10.1126/science.adn0609]

6. Running, L.*, Cristobal, J. R.*, **Karageorgiou, C.**, Camdzic, M., Aguilar, J. M. N., Gokcumen, O., Aga, D. S.#, & Atilla-Gokcumen, G. E.#, 2024. Investigating the mechanism of neurotoxic effects of PFAS in differentiated neuronal cells through transcriptomics and lipidomics analysis. *ACS Chemical Neuroscience*, *15*(24), 4568–4579. [doi: https://doi.org/10.1021/acschemneuro.4c00652]

5. **Karageorgiou, C.**, Gokcumen, O., & Dennis, M. Y.#, 2024. Deciphering the role of structural variation in human evolution: a functional perspective. *Current Opinion in Genetics & Development*, *88*, 102240. [doi: https://doi.org/10.1016/j.gde.2024.102240]

4. Papadadonakis, S., Kioukis, A., **Karageorgiou, C.**, & Pavlidis, P.#, 2024. Evolution of gene regulatory networks by means of selection and random genetic drift. *PeerJ*, *12*, e17918. [doi: https://doi.org/10.7717/peerj.17918]

3. Labbé, F., *et al*., 2023. Genomic analysis of two phlebotomine sand fly vectors of Leishmania from the New and Old World. *PLoS Neglected Tropical Diseases*, *17*(4), e0010862. [doi: https://doi.org/10.1371/journal.pntd.0010862]

2. **Karageorgiou, C.**#, Tarrío, R.#, & Rodríguez-Trelles, F.#, 2020. The cyclically seasonal Drosophila subobscura inversion O7 originated from fragile genomic sites and relocated immunity and metabolic

genes. *Frontiers in Genetics*, *11*, 565836. [doi: https://doi.org/10.3389/fgene.2020.565836]

1. **Karageorgiou, C.**[#], Gámez-Visairas, V., Tarrío, R.[#], & Rodríguez-Trelles, F.[#], 2019. Long-read based assembly and synteny analysis of a reference Drosophila subobscura genome reveals signatures of structural evolution driven by inversions recombination-suppression effects. *BMC Genomics*, *20*(1), 223. [doi: https://doi.org/10.1186/s12864-019-5590-8]

TEACHING & MENTORSHIP

Mentoring:

▫ Carsyn Bonesteel, University at Buffalo, 2024 – Present. BSc Student. Mentoring her independent research project on the evolutionary and functional impact of the hominin-specific *ACOT1* gene duplication on metabolic health.

Awards under supervision:

- Best Idea Award, Department of Biological Sciences, December 2024
- Gerace Dean's Innovation Fund Award, May 2025
- SUNY Summer Undergraduate Research Program (SURP), paid research internship, Summer 2025
- McCroskey Fellowship, October 2025
- UB nominee for the Goldwater Scholarship, December 2025
- GSA Undergraduate Travel Award, January 2026
- Summer Research Experience for Undergraduates (REU), Johns Hopkins University, Summer 2026
- Goldwater Scholarship, March 2026
- UB Excellence in Research, Scholarship and Creativity Award, Spring 2026
- HHMI Cech Fellows Program (Summer Undergraduate Research Program), Summer 2026
- PEQG26 GSA Poster Award Winner

▫ Rebekah Sheih, University at Buffalo, Summer 2024. BSc REU Student. Supervised her REU project on characterizing the evolutionary history of the *ACOT1* locus.
Currently: PhD Candidate at UConn Health.

▫ Eliseo Pampín Bello, Universitat Autònoma de Barcelona, 2017 – 2018. Mentored MSc student and supervised MSc Thesis "Genomic localization and evolutionary history of two aminopeptidase genes long-known to exhibit systematic patterns of spatiotemporal variation in natural populations of *Drosophila"*.

Teaching:

BIO550LEC – Human Evolutionary Genomics, Graduate Course, Department of Biological Sciences, University at Buffalo, Fall 2025. (1 Lecture)

101963 – Genetics, BSc Genetics, Faculty of Biosciences, Universitat Autònoma de Barcelona, 2017 – 2021.
Delivered lectures and served as a teaching assistant for courses covering *Drosophila morphology, Genetic linkage & mapping, Recombination, and Population Genetics*.

APPOINTMENTS & EDITORIAL ROLES

Editor:	Associate Editor for *Heredity* (2025 – Present).
Guest-Editor:	*Heredity*, Special Issue: *Functional and Adaptive Effects of Genomic Structural Variation*, (with Omer Gokcumen, Megan Dennis, and Ellen Leffler). 2024 – 2025. https://www.nature.com/collections/deejeiccee

INVITED SEMINARS

Sept 2026 From the amylase locus to genome-wide structural variation, 3rd Annual University of Utah Rising Stars in Genetics and Genomics Symposium, University of Utah, UT, USA.

July 2026 From the amylase locus to genome-wide structural variation, Department of Molecular Biology and Genetics, Aarhus University, Denmark.

Mar 2026 Why complexity persists: Evolutionary dynamics of the amylase locus in humans and non-human primates, Malaspinas Lab, University of Lausanne, Switzerland.

Mar 2026 Origins of recurrent structural variation: lessons from the primate amylase locus, Flatt Lab, University of Fribourg, Switzerland.

Dec 2025 Origins of recurrent structural variation: lessons from the primate amylase locus, Department of Evolutionary Anthropology, University of Vienna, Austria.

Oct 2025 Why complexity persists: Evolutionary dynamics of the amylase locus in primates, Biology Seminar, iTHEMS, RIKEN, Japan.

Mar 2025 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, Brock University, St Catharines, Canada.

Feb 2024 Ancient *AMY1* gene duplications primed the amylase locus for adaptive evolution upon the onset of agriculture, Biological Sciences Seminar Series, University at Buffalo, NY, USA.

Mar 2022 Comparative population genomics of natural populations of *Drosophila subobscura*: Understanding the establishment and maintenance of chromosomal inversion polymorphisms, The Eric Lai Lab, Sloan Kettering Institute, NY, USA.

Feb 2022 Comparative population genomics of natural populations of *Drosophila subobscura*: Understanding the establishment and maintenance of chromosomal inversion polymorphisms, Gokcumen Lab, University at Buffalo, NY, USA.

Sept 2021 Population genomics of inversion polymorphisms in *Drosophila subobscura*: Mechanisms of establishment and maintenance, Pavlidis Lab, FORTH, Crete, Greece.

CONFERENCE CONTRIBUTIONS

2026 Integrating long-read assemblies and ancestry graphs to map Neanderthal-introgressed structural variation in humans, ***Poster***, ASHG 2026, Montréal, Canada. *Reviewers' Choice Abstract.*

2026 Integrating long-read assemblies and ancestry graphs to map Neanderthal-introgressed structural variation in humans, ***Poster***, PEQG 2026, Pacific Grove, CA, USA.

2025 Convergent evolution through recurrent rearrangements and regulatory rewiring in the primate amylase locus, ***Talk***, ESEB 2025, Barcelona, Spain.

2025 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, ***Poster***, ESEB 2025, Barcelona, Spain.

2025 Why this locus? Recurrent duplications and extensive structural polymorphism at the amylase locus in humans and non-human primates, ***Talk***, Ecological and Evolutionary Genomics, Gordon Research Conference, Renaissance Tuscany Il Ciocco, Italy.

2025 Convergent evolution through recurrent rearrangements in the primate amylase locus, ***Talk***, Evolution 2025, Athens, GA, USA.

2025 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, ***Poster***, Evolution 2025, Athens, GA, USA.

2025 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, ***Talk***, Virtual Evolution 2025, Athens, GA, USA.

2025 Convergent evolution through recurrent rearrangements in the primate amylase locus, ***Talk***, Virtual Evolution 2025, Athens, GA, USA.

2024 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, ***Poster***, Annual Postdoctoral Research Symposium, University at Buffalo, NY, USA.

2024 Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation, ***Talk***, GLAM EvoGen 2024, Syracuse University, NY, USA.

2024 Paleolithic gene duplications primed adaptive evolution of human amylase locus upon agriculture, ***Poster***, SMBE 2024, Puerto Vallarta, Mexico.

2023 Modeling the evolutionary and functional impact of genomic structural variation in mammalian saliva. ***Poster***, Ecological and Evolutionary Genomics, Gordon Research Conference, Bryant University, RI, USA.

2022 Disentangling the origin of the Cretan wild goat *Capra aegagrus cretica*. ***Talk***, 15th International Congress on the Zoogeography and Ecology of Greece and Adjacent Regions, Mytilene, Greece.

2022 Disentangling the origin of the Cretan wild goat *Capra aegagrus cretica*. ***Talk***, Third Joint NEOMATRIX Meeting. Virtual.

2021 The cyclically seasonal *Drosophila subobscura* inversion O_7 originated from fragile genomic sites and relocated immunity and metabolic genes. ***Poster***, SMBE 2021. Virtual.

2021 The cyclically seasonal *Drosophila subobscura* inversion O_7 originated from fragile genomic sites and relocated immunity and metabolic genes. ***Poster***, SMRT Leiden 2021 – Young Investigator Virtual Conference. Virtual.

2019 Long-read based assembly and synteny analysis of a reference *Drosophila subobscura* genome reveals signatures of structural evolution driven by inversions recombination-suppression effects. ***Poster***, VII Jornada de Bioinformatica i Genomica, Barcelona, Spain.

2019 Long-read based assembly and synteny analysis of a reference *Drosophila subobscura* genome reveals signatures of structural evolution driven by inversions recombination-suppression effects. ***Poster***, European Drosophila Research Conference (EDRC), Lausanne, Switzerland.

2019 Comparative population genomics of natural populations of *Drosophila subobscura.* ***Talk***, VII Jornades de Biorecerca UABio. Universitat Autònoma de Barcelona. Barcelona, Spain.

2018 The genome of *Drosophila subobscura*. ***Talk***, VI Jornades de Biorecerca UABio. Universitat Autònoma de Barcelona, Barcelona, Spain.

MEETINGS, ORGANIZATION & OUTREACH

Oct 2026 Moderator, platform abstract session "*Move Over SNPs: Structural Variants, Repeat Expansions, and More*". ASHG 2026, Montréal, Canada.

June 2026 Poster judge for PEQG 2026, Pacific Grove, CA, USA.

May 2026 Moderator, Virtual Evolution 2026 Conference, Session: *Evolutionary Theory*.

August 2025 Outreach Organizer – *Breaking Bread: Exploring the Evolution of Amylase in Humans*. Evolution, Diversity, Art and Community, Festa Major de Gràcia & ESEB 2025, Barcelona, Spain. Organized and led a bilingual (English/Spanish) public outreach

	activity on human evolution and the amylase locus, selected as part of ESEB's outreach initiative. Co-organizer: Thomas Hitchcock.
July 2025	Co-Chair, GRC Power Hour™ – Gordon Research Conference (GRC) on Evolutionary and Ecological Genomics: *Dynamics of Ecological and Evolutionary Change*, Lucca, Italy.
July 2025	Chair for the Gordon Research Seminar (GRS) Ecological and Evolutionary Genomics: *Elucidating the Evolutionary Dynamics of Adaptation in Fluctuating Environments*, Lucca, Italy. Co-chair: Joaquin Nunez.
May 2025	Moderator, Virtual Evolution 2025 Conference, Sessions: *Population Genetics Theory* and *Molecular Evolution*
May 2025	Organization Committee for *Paving Pathways: A multi-panel symposium highlighting scientists in various post-graduate biomedical careers*, Buffalo, USA.
2025 – Present	Organizer of "Talk Nerdy to Me", a monthly event dedicated to public speaking and science communication, aimed at making research accessible to general audiences.
July 2024	Organizer for Society for Molecular Biology and Evolution 2024 Symposium: *Deciphering the functional and adaptive effects of genomic structural variation*, Puerto Vallarta, Mexico. Co-organizers: Megan Dennis, Omer Gokcumen, and Ellen Leffler.
2024 – Present	Organizer, Postdoctoral Meetups, University at Buffalo, USA – Initiated and coordinated regular gatherings to build a supportive community, foster open communication and encourage interdisciplinary exchange.
2022 – Present	Organizer for Genome Structure, Evolution and Anthropology Seminar Series, Buffalo Evolutionary and Anthropological Genomics Lab, University at Buffalo, USA.

SERVICE & MEMBERSHIPS

Reviewer:	Cell, Cell Genomics, Communications Biology, Genetics, Genome Research, Molecular Biology and Evolution, NAR Genomics and Bioinformatics, Nature, Nature Communications, Science Advances, The American Journal of Human Genetics.
Grant Reviewer:	The Leakey Foundation, NordForsk (Nordic Council of Ministers).
Professional Training:	GSA Journals Peer Review Training Program, Winter 2025. eLife Early-Career Reviewers Pool, Spring 2025.
Memberships:	Genetics Society of America (GSA), American Society of Human Genetics (ASHG), Society for Molecular Biology and Evolution (SMBE), Society for the Study of Evolution (SSE), European Society for Evolutionary Biology (ESEB), The Genetics Society (GS).
Committees:	Selection Committee for the Distinguished Postdoc Mentor Award 2025-2026, University at Buffalo, USA.
Departmental Service:	Poster Grader, Biological Sciences Research Symposium 2025, University at Buffalo, USA. Poster Grader, Biological Sciences Research Symposium 2026, University at Buffalo, USA.

MEDIA HIGHLIGHTS & PRESS COVERAGE

2026	Invited commentary on "*Archaeogenetic insights into the demographic history of Late*

	Neanderthals" (PNAS), featured in *National Geographic*.
2026	*Heredity* podcast guest, discussing the Special Issue "*Functional and Adaptive Effects of Genomic Structural Variation*".
2026	*Heredity* podcast guest, discussion of the journal's new ECR Editor training program.
2024	Yilmaz, Karageorgiou et al. Reconstruction of the human amylase locus reveals ancient duplications seeding modern-day variation. *Press release*. Highlights: *NYT*, *Discover*, *CNN*, *NBC*. *Highlighted in the NIH Director's blog, as review highlight in Cell Genomics, PLOS SciComm "Top Stories in Human Evolution of 2024", Leakey Foundation Podcast "Top Human Origins Discoveries of 2024".*

PROGRAMMING LANGUAGES

Bash, Python, R, MySQL, HTML, Perl

REFERENCES

► Omer Gokcumen <omergokc@buffalo.edu> ; *Professor*, *University at Buffalo* ; Postdoc Mentor.

► Charles Lee <Charles.Lee@jax.org> ; *The Robert Alvine Family Chair and Professor, The Jackson Laboratory for Genomic Medicine* ; Collaborator and co-author.

► Megan Dennis <mydennis@ucdavis.edu> ; *Professor, Biochemistry & Molecular Medicine, M.I.N.D. Institute, Genome Center, University of California Davis* ; Collaborator and co-author.

► Stefan Ruhl <shruhl@buffalo.edu> ; *Professor and Chair of Oral Biology, University at Buffalo School of Dental Medicine* ; Collaborator and co-author.

Updated November 2026